Renyi Yang

renyi.yang@grenoble-inp.org [mail] (+33) 06 51 67 12 78
Github: Renyi Yang [web] Personal Website [web]

EDUCATION

ShanghaiTech University [web] **Shanghai, CN**
B.ENG. in Computer Science and Technology *Sep. 2023 - Current*

- GPA: 3.82/4.0 — Major GPA: 4.0/4.0
- Rank (CS): 14/171 — Rank (CS + EE): 21/266

Grenoble INP [web] **Grenoble, FR**
Signal Image Processing Methods and Applications *Jan. 2026 - Current*

INTERNSHIP

Advanced Micro Devices, Inc. [web] *Sep. 2025 - Jan. 2026*
AI Software Engineer Intern

- Developed **HeteroOCR**, an CPU/GPU/NPU-optimized OCR system for complex documents.
- Showcased at **CIIE 2025** as AMD's flagship NPU-AI demo.

Shanghai Azimuth Data Technology Co., Ltd. [web] *Jul. 2025 - Sep. 2025*
Software Engineer Intern

- Developed a **multi-COM** GNSS client that concurrently parses **NMEA 0183** and **RTCM 3.3** streams
- Built a **LAN-based** simulator client to both ingest and transmit messages

RESEARCH EXPERIENCE

Post-Moore Microelectronics and Integrated Circuit Center (PMICC) [web] *Jul. 2025 - Current*
Research Assistant, advised by Dr. Yajun Ha

- Accelerated ResNet50 inference on an NPU using **MLIR-AIE**
  - Mapped compute/dataflow, optimized tensor tiling and on-chip buffer reuse
- Developing a **high-throughput Chinese OCR** pipeline for heterogeneous case files
  - Layout, Detection, Recognition

Network Intelligence Center (NICE) [web] *Jul. 2024 - Jun. 2025*
Research Assistant, advised by Dr. Dingzhu Wen

- Implemented and evaluated FL algorithms optimized for **resource-constrained** environments

COURSE PROJECTS

Network on acoustic link [code] *Sep. 2025 - Dec. 2025*
Team Leader

A **Go**-based acoustic communication system
- Physical, MAC, Network, Transport, Application layers

Oat Compiler [opt] [type-check] *Sep. 2025 - Dec. 2025*
Team Leader

A simple **compiler** written in **OCaml**
- Lexing, Parsing → LLVM → Assembly, with Type Checking, Optimization

Pintos [code] *Mar. 2025 - Jun. 2025*
Team Leader

A simple **operating system** framework
- Threads, User Program + Syscalls, Virtual Memory and File System

Plants vs. Zombies [code] *May 2024 - Jun. 2024*
Personal Project

A PvZ game based on **C++** with **CMake**

VOLUNTEERING

Volunteer for summer and winter activities [web] *June 2023 - Aug. 2024*
Teaching Assistant

136 Hours in total, providing lessons using specialized teaching methods and materials

AWARDS & HONORS

- **AMD Open Hardware 2025 Competition** [web]: HeteroOCR

TECHNICAL STRENGTHS

Programming Languages	Go, C&C++, Ocaml, Python, Matlab, RISC-V, X86lite
Framework & Toolchain	Git, Linux, Docker, Conda, Vim, GNU Make, CMake
Misc	LaTeX, Markdown, Marp